STAFF REVIEW · H446 302 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Requirement for an algorithm to work correctly.

2. To expose faults on boundaries/unusual inputs.

3. A property that stays true after each loop iteration.

4. Indexes 0 and 2 sum8; distinct values {4,1,2} count3.

I Do — source worked steps

1. Predict first — Duplicates do not increment the counter because each equals the preceding element.

2. Explain the method — Only changes to five and seven increment count; result is three distinct values.

We Do — guided CHECK

1. Predict first
Three initial ones produce one distinct group; 3 and 4 add two groups.
Explain the method
Test empty input and an all-equal array to confirm base conditions and comparisons.

You Do — source mark schemes

1. Start count1 [1]; i1 still1 [1]; i2 still1 [1]; i3→2 [1]; i4→3 [1]; i5 remains3 [1].

2. Empty []→0 [2]; equal [5,5]→1 [2]; distinct [1,2,3]→3 [2].

3. Algorithm counts transitions and reports3 rather than2 [2]; pre-sort then scan [2]; sorting adds typical O(n log n) time [1], or use set/hash with expected O(n) and space O(n) [1].

Exit Ticket — source answer

1. False. Equal values might be separated and overcounted.